DETECTION OF FAKE AND AI-GENERATED PRODUCT REVIEWS USING DEEP LEARNING: A COMPARATIVE STUDY OF FOUR ARCHITECTURE FAMILIES

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Detection of Fake and AI-Generated Product Reviews Using Deep Learning: A Comparative Study of Four Architecture Families

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***Abstract*—Large language models can now write fluent product reviews at almost no cost, and people cannot reliably tell them from reviews written by customers. We compare four deep learning architecture families on separating computer-generated reviews from original ones: a TF-IDF + MLP baseline, a Text-CNN, a BiLSTM + Attention model and a fine-tuned DistilBERT. All four use the Fake Reviews Dataset of Salminen et al., about 20,000 GPT-2 reviews and 20,000 original Amazon reviews, with one stratified 70/15/15 split and one evaluation harness. On 6,074 test reviews DistilBERT is the most accurate, at 97.09% accuracy, 0.9708 macro F1 and 0.9980 ROC-AUC. BiLSTM + Attention reaches 95.55%, TF-IDF + MLP 95.03% and Text-CNN 92.94%. DistilBERT makes 41% fewer errors than TF-IDF + MLP but has thirteen times as many parameters. All four models are least accurate on reviews of 20 words or fewer. The error analysis also exposes a shortcut. In the test set, 64.5% of computer-generated reviews end without closing punctuation, against 9.2% of original ones, and every model is more accurate on computer-generated reviews that end this way. The planned test on human-written hotel reviews from another domain was not carried out.**

***Keywords*—fake review detection, machine-generated text, text classification, DistilBERT, bidirectional LSTM, attention, convolutional neural network, TF-IDF**

I. Introduction

Product reviews guide online purchases only as long as customers wrote them. Review farms have broken that assumption for years. Large language models now break it at almost no cost, because they write fluent, product-specific text on demand. Human raters in the study of Salminen et al. [1] identified GPT-2 reviews with 55.36% accuracy, barely above chance. Detection therefore has to be automatic.

We treat it as binary text classification, in which each review is either original or computer-generated. Published detectors cannot be ranked against each other, since each one reports results on its own dataset, split and metrics. That leaves a practical question open: how much does a large pre-trained model add over a cheap one when everything else is equal? This project answers it for four architecture families and records parameter count, training time and inference latency next to accuracy.

The study covers English product reviews from the Fake Reviews Dataset [1] and models small enough to train on one GPU or a laptop CPU. Section II reviews prior work, Section III states the objectives, Sections IV and V describe the models and the shared setup, and Section VI reports and discusses the results.

II. Literature Review

Early detectors relied on hand-built features. Jindal and Liu [2] trained logistic regression on review, reviewer and product features from 5.8 million Amazon reviews. They reached 98.7% AUC on two types of spam, yet text features alone gave 63%. Ott et al. [3] collected 400 truthful hotel reviews and 400 deceptive ones written by crowd workers. A linear SVM on psycholinguistic and bigram features scored 89.8% accuracy, while human judges scored between 53% and 62%. The corpus was later extended with negative reviews [4], and the extended corpus was our intended cross-domain test set.

Neural networks learn their features from the text instead. Kim [5] showed that one convolutional layer with filters of different widths over word vectors performs well on seven sentence classification tasks, and our Text-CNN follows that design. Long short-term memory networks [6] carry information across long word sequences, and attention, introduced for machine translation [7], lets a network weight the positions that matter most. Ren and Zhang [8] applied both ideas to deceptive opinion spam. Their CNN builds sentence vectors that a gated recurrent network with attention combines into one document vector, and it beat feature-based methods on hotel, restaurant and doctor reviews. BiLSTM + Attention in this study is a simpler model of the same kind.

The Transformer [9] and the pre-training methods built on it changed the picture again. BERT [10] is pre-trained on unlabelled text and then fine-tuned for a target task. DistilBERT [11] compresses it into a model that is 40% smaller and 60% faster and keeps 97% of its language understanding. Fine-tuned Transformers now hold the best published results for fake review detection. Salminen et al. [1] detected their GPT-2 reviews with a fine-tuned RoBERTa at 97% precision and recall. AlQadi et al. [12] reached 97.25% accuracy with DistilBERT on 2,000 Yelp reviews, half of them rephrased by ChatGPT. Geetha et al. [13] report 98% accuracy for a DeBERTa model tuned with Monarch Butterfly Optimization, on three datasets that include ours.

Other detectors depend on information that a review platform seldom has. DetectGPT [14] needs no labelled data, but it scores text with the log probabilities of the model that wrote it, and it reached 0.95 AUROC on news articles from GPT-NeoX. FraudSquad [15] joins language-model embeddings with a graph transformer over reviewers and products and scores from 89.45% to 99.98% on five datasets, but it cannot run without the review graph.

Each study above evaluates one main model on its own data, and the datasets differ in size, domain and generator. None of them measures what a smaller or older architecture loses under identical conditions. That comparison is the subject of this report.

III. Objectives

1) To quantify the detection performance of four architecture families, namely feed-forward, convolutional, recurrent and self-attention, on fake and AI-generated product reviews under one common evaluation protocol.
2) To characterise the trade-off between detection accuracy and computational cost, measured as parameter count, training time and inference latency, for each of the four architectures.
3) To determine how far detectors trained on machine-generated product reviews keep their accuracy on human-written deceptive reviews from a different domain.

IV. Methodology

The four models differ only in how they turn a review into a prediction. They share the cleaned data, the split and the evaluation harness described in Section V. Each model fits its own vocabulary or features on the training split and outputs the probability that a review is computer-generated, and a probability of 0.5 or more counts as a computer-generated prediction. Table I lists the settings of all four.

Table I
Model Settings

Model	Input	Main layers	Training	Epochs	Params
TF-IDF + MLP	Word 1-2-grams (15,000) and character 3-5-grams (25,000)	Dense 128 and 64, ReLU	Adam, lr 1e-3, L2 1e-4, batch 256	8	5,128,449
Text-CNN	Word tokens, vocabulary 20,000, length 128	Embedding 100, convolutions of width 3, 4, 5 with 100 filters, max-pool, dropout 0.5	Adam, lr 1e-3, batch 64	5	2,120,902
BiLSTM + Attention	Word and punctuation tokens, vocabulary 30,000, length 256	Embedding 128, BiLSTM 2 x 128, attention pooling, dropout 0.5	Adam, lr 1e-3, batch 64, clip 5, early stopping	max 10	4,104,706
DistilBERT	WordPiece subwords, length 128	Pre-trained DistilBERT encoder, two-layer classification head	AdamW, lr 2e-5, decay 0.01, batch 32, warm-up then linear decay	3	66,955,010

A. TF-IDF + MLP (Vedant Agarwal)

TF-IDF + MLP represents a review only by the words and character sequences it contains. Two scikit-learn [16] vectorisers are fitted on the training reviews. One keeps word unigrams and bigrams that appear in at least two reviews, while the other keeps sequences of three to five characters that appear in at least five. Both weight terms by sublinear TF-IDF, and their outputs are joined into one vector of 40,000 features. A multilayer perceptron with two hidden ReLU layers of 128 and 64 units maps this vector to a single output and is trained with Adam and an L2 penalty for eight epochs.

The model ignores word order beyond two-word phrases. Its accuracy therefore shows how much of the task surface statistics can solve. Character features were included because they record spelling, punctuation and parts of words that a word vocabulary drops.

B. Text-CNN (Ishan Satyanand Thakur)

Text-CNN follows the sentence CNN of Kim [5] and looks for short phrase patterns anywhere in a review. Its tokeniser lower-cases the text and keeps runs of word characters, so punctuation is removed. The 19,998 most frequent training tokens, a padding symbol and an unknown symbol form a vocabulary of 20,000 entries, and each review becomes a sequence of 128 token indices. A 100-dimensional embedding layer, trained from random initial values, feeds three parallel convolutions with windows of three, four and five tokens and 100 filters each. Every feature map passes through ReLU and is max-pooled over the whole sequence, which keeps the strongest response of each filter but discards where it occurred. The three pooled vectors are joined, passed through dropout and mapped to two classes.

The network is trained with cross-entropy and Adam for five epochs, and the weights after the fifth epoch are used. It was chosen as the cheapest neural model in the comparison, one that can learn stock phrases such as generic praise without modelling the whole review.

C. BiLSTM + Attention (Vedant Totla)

BiLSTM + Attention reads each review in order, in both directions. Its tokeniser lower-cases the text and keeps words, numbers and single punctuation marks as separate tokens. A vocabulary of 30,000 entries is built from the training reviews in the same way as for Text-CNN, and each review is cut or padded to 256 tokens. A 128-dimensional embedding feeds a bidirectional LSTM [6] with 128 units per direction. For attention pooling [7], a linear layer scores the tanh of each hidden

state. Padding positions are masked, a softmax turns the scores into weights, and the weighted sum of hidden states goes to one output unit. Dropout is applied to the embeddings and to this pooled vector.

Training minimises binary cross-entropy with Adam, clips gradients at norm 5 and runs for at most ten epochs. After each epoch the model is scored on the validation split. The weights with the best validation F1 are kept, and training stops after two epochs without improvement. The model was chosen because it uses word order and long context, and its attention weights show which tokens drove a prediction, which the error analysis uses.

D. DistilBERT (Rudra Rajpurohit)

DistilBERT fine-tunes the pre-trained distilbert-base-uncased checkpoint [11] with the Hugging Face Transformers library [17]. Its WordPiece tokenizer lower-cases each review and splits it into subword units. Reviews are cut to 128 tokens and padded within each batch. A randomly initialised classification head of two linear layers sits on the encoder output. All 66,955,010 parameters are updated with AdamW for three epochs. The learning rate rises linearly over the first 10% of steps and then decays linearly to zero, and gradients are clipped at norm 1.0.

DistilBERT represents the pre-trained Transformer family, which holds the best published results on this task [1], [12], [13]. It was preferred to full BERT because it is 40% smaller [11], which keeps fine-tuning within a single session on a free cloud GPU.

V. Experimental Setup

A. Dataset

The data are the Fake Reviews Dataset of Salminen et al. [1], taken from the Kaggle copy of its OSF release. It contains 40,526 product reviews, each with a category, a star rating, the review text and a label. Label 1 marks a computer-generated review, written by a GPT-2 model trained on Amazon reviews, and label 0 marks an original Amazon review. The classes are balanced at 20,294 computer-generated and 20,232 original reviews.

Fig. 1 shows the main distributions after cleaning. Ten categories hold between 3,585 and 4,727 reviews each, and eight minor categories hold 64 reviews between them. Star ratings are almost identical across the classes, with means of 4.26 for computer-generated and 4.25 for original reviews, so the rating carries no label information. Length differs more. Computer-generated reviews average 61.2 words against 73.6 for original ones, with medians of 35 and 42.

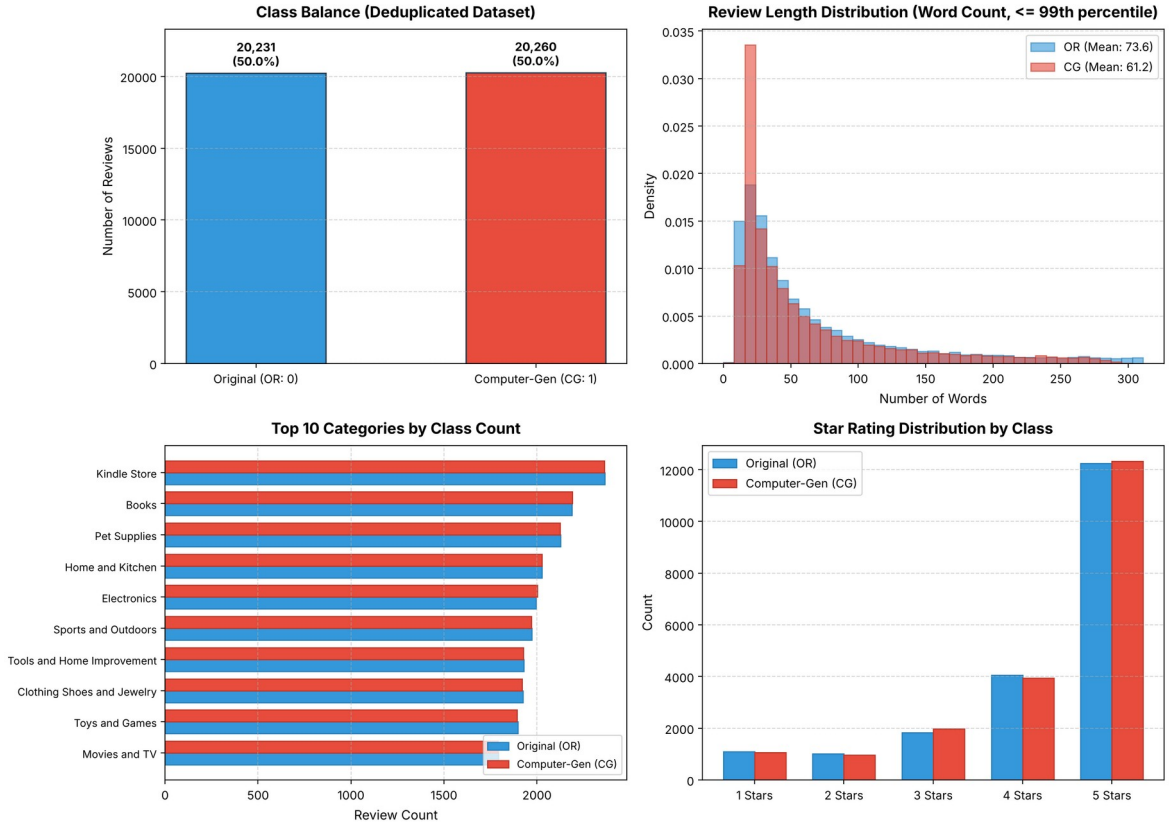


Fig. 1. Dataset distributions after cleaning: class balance, review length in words, the ten main categories and star ratings, for original (OR) and computer-generated (CG) reviews.

B. Preprocessing and Split

One shared loader prepares the data for every model. It drops rows with a missing text or label, strips surrounding whitespace and removes exact duplicate texts, so that no duplicate can sit in both the training and the test split. Cleaning removed 35 rows and left 40,491 reviews, 20,260 computer-generated and 20,231 original. A stratified 70/15/15 split with random seed 42 then gives 28,343 training, 6,074 validation and 6,074 test reviews. The validation and test splits each

contain 3,039 computer-generated and 3,035 original reviews. Vocabularies and features are fitted on the training split, model selection uses the validation split, and the test split is read once for the reported numbers.

C. Evaluation Metrics

All models report through one evaluation harness built on scikit-learn [16]. It computes accuracy, macro F1, ROC-AUC, and precision and recall with the computer-generated class as positive. ROC-AUC uses the predicted probabilities, and the other metrics use the 0.5 threshold. The harness also records three costs: the number of trainable parameters, the wall-clock training time and the inference latency, which is the mean time per review over three runs on 64 test reviews after a warm-up batch. For the neural models, training time includes the validation pass after each epoch. For TF-IDF + MLP it includes fitting the vectorisers.

D. Implementation and Hardware

The neural models are written in PyTorch [18]. TF-IDF + MLP and BiLSTM + Attention were trained on a CPU, Text-CNN on a CUDA GPU and DistilBERT on a Tesla T4 GPU in Google Colab. Training time and latency are therefore comparable only between models on the same kind of device. Every model was trained once. The code, the results and the commit history are public at https://github.com/RajRudra06/DL_Project.

VI. Results and Discussion

A. Overall Comparison

Table II gives the test results. DistilBERT is the most accurate model at 97.09%, and it also leads on recall, macro F1 and ROC-AUC. TF-IDF + MLP has the highest precision, 0.9563. By macro F1 the order is DistilBERT, BiLSTM + Attention, TF-IDF + MLP and Text-CNN. The gaps are uneven. DistilBERT leads BiLSTM + Attention by 1.54 points of accuracy, BiLSTM + Attention leads TF-IDF + MLP by 0.52 and TF-IDF + MLP leads Text-CNN by 2.09. Each model was trained once, so the 0.52-point gap may not hold for another seed. Every ROC-AUC is at least 0.981.

Table II
Test Results (6,074 Reviews)

Model	Acc.	Prec.	Rec.	Macro-F1	ROC-AUC	Params	Train (s)	Latency (ms)	Device
TF-IDF + MLP	0.9503	0.9563	0.9437	0.9503	0.9904	5.13M	240	0.71	CPU
Text-CNN	0.9294	0.9525	0.9039	0.9293	0.9811	2.12M	136	0.08	GPU
BiLSTM + Attention	0.9555	0.9416	0.9714	0.9555	0.9920	4.10M	1,126	0.95	CPU
DistilBERT	0.9709	0.9520	0.9918	0.9708	0.9980	66.96M	930	3.59	GPU (T4)

DistilBERT is also the most expensive model. It has 66.96 million parameters, thirteen times as many as TF-IDF + MLP and sixteen times as many as BiLSTM + Attention, and it needs 3.59 ms per review on a GPU. TF-IDF + MLP needs 0.71 ms per review on a CPU and trains in four minutes. In return, DistilBERT makes 177 errors on the test set against 302 for TF-IDF + MLP, 41% fewer.

B. Error Types

The models fail in different directions, as Table III shows. Text-CNN misses 292 of the 3,039 computer-generated test reviews, which gives it the lowest recall at 0.904. BiLSTM + Attention and DistilBERT miss only 87 and 25, but they flag more original reviews as computer-generated, 183 and 152. TF-IDF + MLP splits its 302 errors nearly evenly.

Table III
Error Counts on the Test Set

Model	Original reviews flagged as computer-generated	Computer-generated reviews missed	Total errors
TF-IDF + MLP	131	171	302
Text-CNN	137	292	429
BiLSTM + Attention	183	87	270
DistilBERT	152	25	177

C. Error Analysis

Short reviews are the hardest for every model. Table IV splits the test set by length. On reviews of 20 words or fewer, accuracy ranges from 0.843 for Text-CNN to 0.931 for DistilBERT. Above 50 words all four models exceed 0.96.

Table IV
Accuracy by Review Length in Words

Model	<= 20 (1,467 reviews)	21-50 (2,110)	51-100 (1,205)	> 100 (1,292)
TF-IDF + MLP	0.879	0.958	0.988	0.983
Text-CNN	0.843	0.946	0.972	0.961
BiLSTM + Attention	0.889	0.965	0.987	0.987
DistilBERT	0.931	0.976	0.993	0.987

Review endings carry a strong cue. Of the 3,039 computer-generated test reviews, 1,960 (64.5%) end without closing punctuation, defined as a full stop, question mark, exclamation mark, closing quote or closing bracket. Only 280 of the 3,035 original reviews (9.2%) do. Table V shows that every model is more accurate on computer-generated reviews

without closing punctuation than on those with it. The gap ranges from 1.2 points for DistilBERT to 4.5 points for BiLSTM + Attention.

TABLE V
ACCURACY ON COMPUTER-GENERATED TEST REVIEWS BY ENDING

Model	No closing punctuation (1,960)	Closing punctuation (1,079)	Gap (points)
TF-IDF + MLP	0.951	0.931	1.9
Text-CNN	0.915	0.884	3.1
BiLSTM + Attention	0.987	0.943	4.5
DistilBERT	0.996	0.984	1.2

The models also fail on largely different reviews. Only 38 test reviews defeat all four, 26 original and 12 computer-generated, with a mean length of 17 words. A majority vote of the four models scores 0.9659, below DistilBERT alone. Table VI lists these and the other supporting checks.

TABLE VI
SUPPORTING ANALYSES

Analysis	Why it was run	Result	What it shows
Reviews wrong for all four models	Find inputs no model handles	38 reviews: 26 original, 12 computer-generated, mean 17 words	Shared failures are short reviews
Majority vote of the four models	Test whether combining helps	Accuracy 0.9659	Below DistilBERT alone (0.9709)
Pairwise disagreement	Measure how far errors overlap	269 reviews (BiLSTM + Attention vs DistilBERT) to 476 (Text-CNN vs DistilBERT)	Errors overlap only in part
Validation against test accuracy	Check for overfitting to validation	DistilBERT 0.9714 vs 0.9709, TF-IDF + MLP 0.9475 vs 0.9503	Validation and test agree
BiLSTM + Attention early stopping	Check when training stopped	All 10 epochs ran, best validation F1 0.9597	Early stopping never triggered

Single reviews show the same patterns. DistilBERT gave a probability of 0.9997 that the original review "This novel is a masterpiece. The plot is gripping and the characters are well-developed." was computer-generated. BiLSTM + Attention gave 0.811 for the original review "this gown is sheer. It is beautiful and very comfortable", with its highest attention weights on "comfortable", "very" and "beautiful". The same model called a computer-generated review that repeats "it arrived on the same day" over and over original, at 0.075, with attention on "same" and "day". For a correctly classified negative original review, attention fell on "!", "?" and "trash".

D. Discussion

Pre-training is the most likely reason for the lead of DistilBERT. Its encoder has already learned how fluent English is distributed, so the repetitive or incoherent stretches that GPT-2 sometimes produces stand out without having to be learned from 28,343 examples. TF-IDF + MLP comes within about two points because computer-generated text also leaves surface traces, such as repeated phrases and unfinished sentences, that character n-grams record without any notion of meaning. BiLSTM + Attention adds word order and reads 256 tokens, twice as many as Text-CNN and DistilBERT. Its gain over TF-IDF + MLP is small enough that one seed cannot confirm it.

Text-CNN has a more specific weakness. Its tokeniser removes punctuation and its pooling discards position, so it cannot see whether a review stops in mid-sentence. It was also trained for a fixed five epochs from random embeddings, whereas BiLSTM + Attention kept its best validation weights and DistilBERT started from pre-trained ones. Either cause fits the many computer-generated reviews it misses, but we ran no ablation, so their shares are unknown.

Short reviews are hard because they hold little evidence. Two sentences of generic praise look the same whether a customer or GPT-2 wrote them, and the examples above show mistakes in both directions on exactly such text. The 38 reviews that defeat all four models average 17 words, which fits this explanation.

The ending cue has the widest consequences. Many GPT-2 reviews in this dataset stop in mid-sentence, which suggests that generation was cut at a length limit. Every model gains from the cue, so part of the reported accuracy measures an artefact of how the dataset was built and not the style of machine-written text. A modern generator that finishes its sentences would remove the cue, and we expect accuracy on its reviews to be lower than Table II suggests.

For a review platform the choice depends on the cost of a mistake. TF-IDF + MLP gives 95% accuracy for a small share of the compute and suits bulk screening, while DistilBERT suits cases where each missed fake is expensive. Combining the four does not help, since the three weaker models can outvote DistilBERT on reviews that it alone classifies correctly.

E. Comparison With Prior Work and Status of the Objectives

The 97.09% accuracy of DistilBERT is close to the 97% precision and recall that Salminen et al. [1] report for RoBERTa. It is somewhat below the 98% that Geetha et al. [13] report for a tuned DeBERTa on data that includes this dataset. Their splits and preprocessing differ from ours, so the comparison is indicative only. Objectives 1 and 2 were met. Objective 3 was not. The trained models were never applied to the hotel corpus of Ott et al. [3], [4], so this study cannot say how far they transfer to human-written deceptive reviews.

VII. Conclusion

With the data, split and metrics held fixed, a fine-tuned DistilBERT separates GPT-2 product reviews from original ones best, at 97.09% accuracy and 0.9980 ROC-AUC. BiLSTM + Attention and TF-IDF + MLP follow at 95.55% and 95.03%, and Text-CNN trails at 92.94%. Pre-training buys about two points of accuracy and 41% fewer errors than the TF-IDF baseline, at thirteen times the parameters. All four models struggle with short reviews, and all four profit from the unfinished endings of GPT-2 text, which limits what these numbers say about fakes from newer generators.

VIII. Future Work

The missing cross-domain test on the corpus of Ott et al. [3], [4] comes first, since it would show how much of the learned signal is specific to GPT-2. The ending cue should then be removed, either by trimming unfinished final sentences or by testing only on reviews that end cleanly, and reviews from recent instruction-tuned models should be added. The comparison itself can be made firmer. Each model should be trained with different seeds and tuned on the validation set, Text-CNN should get validation-based model selection and pre-trained embeddings, and all four should be timed on one machine. For deployment, quantisation or distillation into a smaller model could cut the cost of DistilBERT.

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Use of AI Tools

Claude (Anthropic) was used to write scaffolding code for the shared evaluation pipeline and the DistilBERT training script, and to help draft parts of this report. All experiments were run by the team.

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C.3 : Individual Contribution Statement

Member Name	Reg. No.	Model / Task Owned	Key Contributions	Signature
Rudra Rajpurohit	230911570	DistilBERT. Evaluation harness, metrics and comparison table	Set up the GitHub repository. Built the shared evaluation harness (evaluate.py), the data loader and split (data.py) and the comparison script (compare.py). Fine-tuned DistilBERT on a Colab T4 GPU and analysed its errors. Researched and compiled the literature review. Wrote the nextSteps.md hand-off file for the team.	
Vedant Totla	230911526	BiLSTM + Attention	Built BiLSTM + Attention (bilstm_model.py), with a vocabulary fitted on the training split, validation-based early stopping, test evaluation and export of attention weights for the error analysis.	
Ishan Satyanand Thakur	230953356	Text-CNN. Tokenisation and embeddings	Built Text-CNN (textcnn_model.py) and the tokeniser and embedding utilities (preprocess.py). Added the model results to the repository README and updated nextSteps.md.	
Vedant Agarwal	230953312	TF-IDF + MLP. Exploratory data analysis	Built TF-IDF + MLP (tfidf_mlp_model.py). Wrote the exploratory data analysis script (eda.py), which produces the dataset summary tables and Fig. 1.	